BLEEP Education – Artificial Intelligence Internship
June Batch

EMPLOYEE ATTRITION & WORKFORCE ANALYTICS

Exploratory Data Analysis

Presented by: Tripti

Dataset: IBM HR Analytics Employee Attrition & Performance Dataset

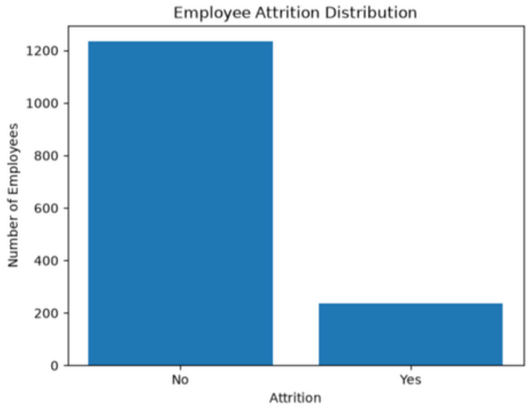
PROJECT OBJECTIVE

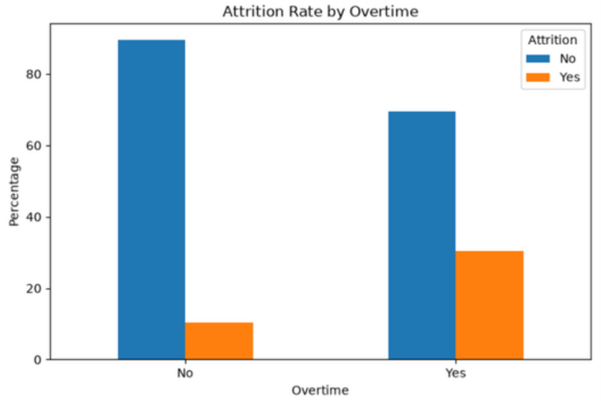
- Analyze employee workforce data
- Understand factors associated with employee attrition
- Identify patterns in employee demographics and workplace conditions
- Generate HR-oriented insights for employee retention

DATASET & DATA UNDERSTANDING

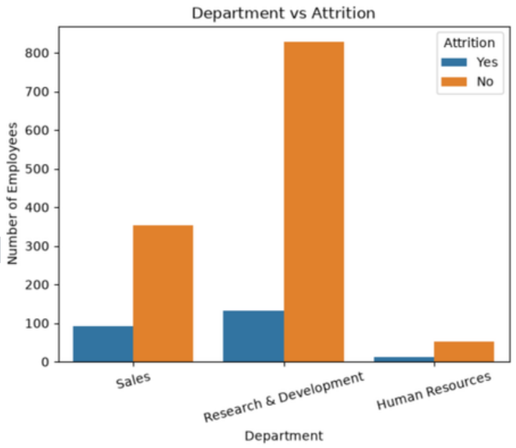
- 1,470 employees
- 35 features
- 26 numerical + 9 categorical
- Target: Attrition
- No → 1,233 employees
- Yes → 237 employees
- No missing values
- Key features: Age, Monthly Income, Job Role, Job Satisfaction, Overtime, Distance From Home, Years at Company

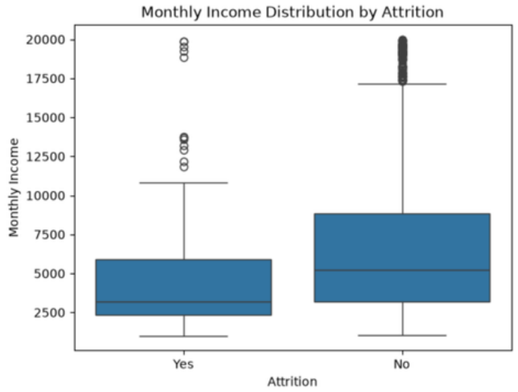
DATA ANALYSIS & EDA

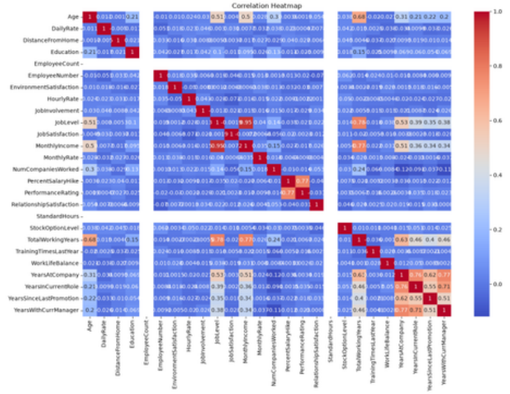












These are highlights, not the complete EDA.

KEY FINDINGS

- Attrition is unevenly distributed across employees
- Overtime shows a noticeable relationship with attrition
- Job satisfaction differs across attrition outcomes
- Attrition patterns vary across departments
- Age and monthly income distributions differ between groups
- Distance from home and years at company provide additional retention-related patterns

CONCLUSION & HR INSIGHTS

- EDA helps identify workforce patterns that can support employee-retention analysis.
- Workload and overtime should be investigated
- Employee satisfaction can be monitored
- Compensation and career progression are relevant areas
- Department-level differences can guide further analysis
- The findings provide a foundation for future predictive modeling

THANK YOU